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9. *(a)* (i) The average bond enthalpy of a C—C bond is quoted as 348 kJ mol^{-1}.

Explain what is meant by *bond enthalpy*. [2]

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(ii) Ethyne, C_2H_2, contains a C≡C. It reacts with hydrogen in a similar way to ethene.

$$H{-}C{\equiv}C{-}H \quad + \quad 2H_2 \longrightarrow C_2H_6$$

Some average bond enthalpies are given in the table.

Bond	Average bond enthalpy / kJ mol^{-1}
C≡C	839
C—C	348
C—H	413
H—H	436

Use the data to calculate the enthalpy change, ΔH, for the reaction of ethyne and hydrogen. [3]

ΔH = ... kJ mol^{-1}

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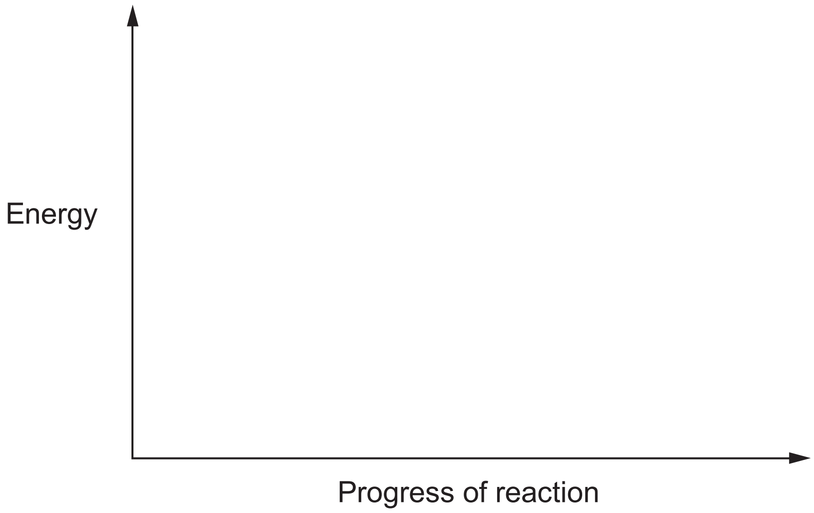
(iii) Use your answer to part (ii) to sketch an energy profile diagram for this reaction on the axes below. Label ΔH and the activation energy, E_a, on your diagram. [2]



(b) Enthalpy changes of reaction are often found indirectly. The enthalpy change for the reaction of ethyne with hydrogen, as shown in part *(a)*, can be determined by using enthalpy changes of combustion.

The table gives some enthalpy changes of combustion, $\Delta_c H^\theta$.

Substance	Enthalpy change of combustion, $\Delta_c H^\theta$ / kJ mol^{-1}
hydrogen, H_2	–286
ethyne, C_2H_2	–1300
ethane, C_2H_6	–1600

Use these enthalpy changes to calculate the enthalpy change, ΔH, for the reaction of ethyne and hydrogen. [3]

$$C_2H_2 \quad + \quad 2H_2 \longrightarrow C_2H_6$$

ΔH = ... kJ mol^{-1}

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(c) The theoretical values that you have calculated in parts *(a)*(ii) and *(b)* are both the enthalpy change for the reaction between ethyne and hydrogen.

Suggest a reason why these values are not the same. [1]

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(d) Suggest the type of reaction that occurs between ethyne and hydrogen. [1]

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